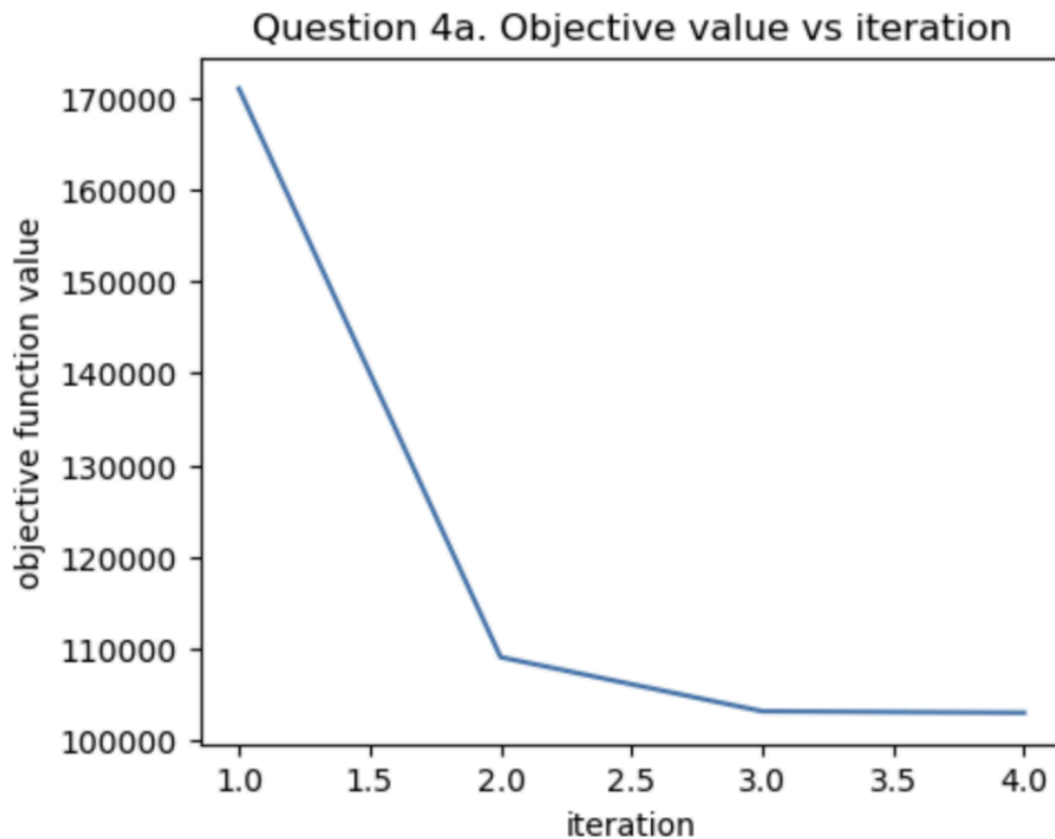


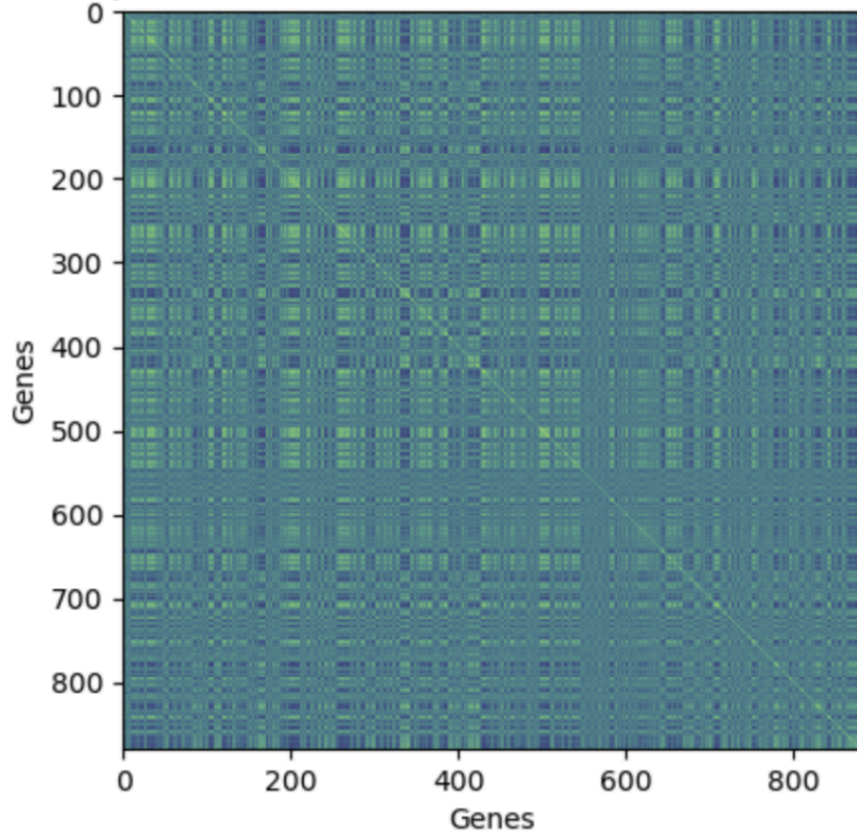
(a) Run K-means algorithm with $K = 3$ with the initialization of K means given in 'test mean.txt'. Plot the objective (y-axis) over iterations (x-axis). What is the objective value at convergence?



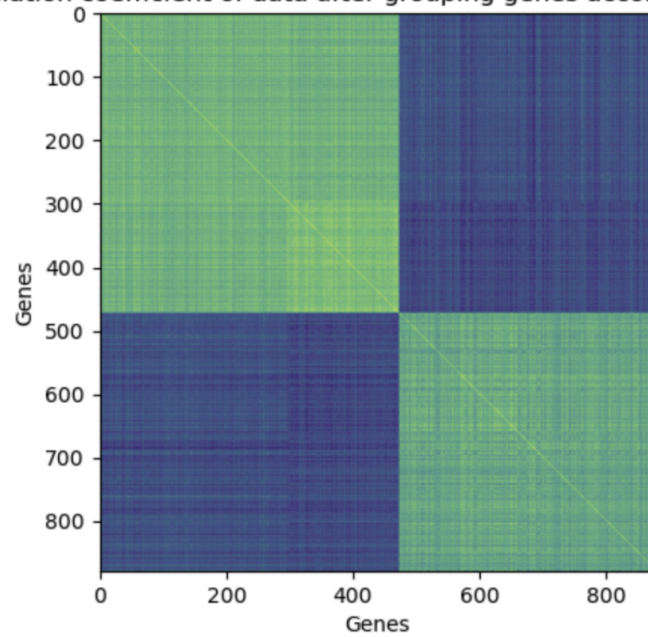
4a answer: the objective value at convergence is 103009.47330276546

(b) Run K-means algorithm with $K = 3$ with your own random initialization. Plot the correlation coefficient matrix of a) the raw data and b) the data after you group the columns of the expression matrix according to the clusters found by K-means. Plot these as an image of $N \times N$ correlation matrix for N genes.

Question 4b. Correlation coefficient of raw data

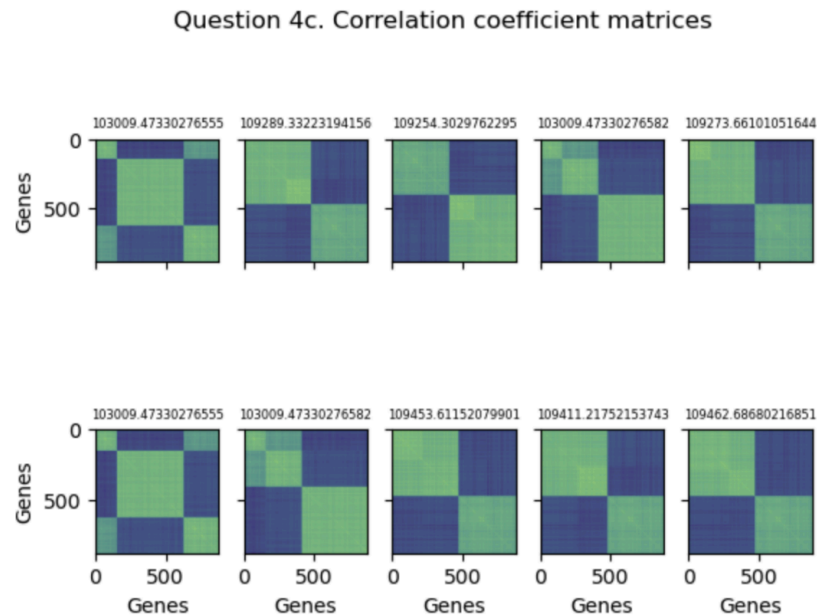


Question 4b. Correlation coefficient of data after grouping genes according to k means clustering



(c) With $K = 3$, try 10 random initializations. Show the correlation coefficient matrix as above and the corresponding objective at convergence for each initialization. Which of the 10 results would you choose for follow-up analysis?

I would choose the initializations which give the least objective function value for follow-up analysis. In this case, it would be the first and sixth initializations (sixth is bottom row first matrix).



(d) Run K-means algorithm for $K = 3, \dots, 12$, with 10 random initializations for each K . Show the correlation coefficient matrix and objectives at convergence for different K , with your best choice of the 10 runs for each K . What do you think is the best K ? Why?

Looking at a plot of objective value vs K-value (see python notebook for this plot, not shown here) of the obtained results and considering the elbow method of choosing best K, I don't see any abrupt decrease in objective value. Therefore, it is difficult to choose the optimal K. However, the decrease at $k = 4$ looks slightly promising as the drop in the plot is slightly steeper than other drops. So, by this logic I would choose the best K as 4. Below are the correlation matrices.

